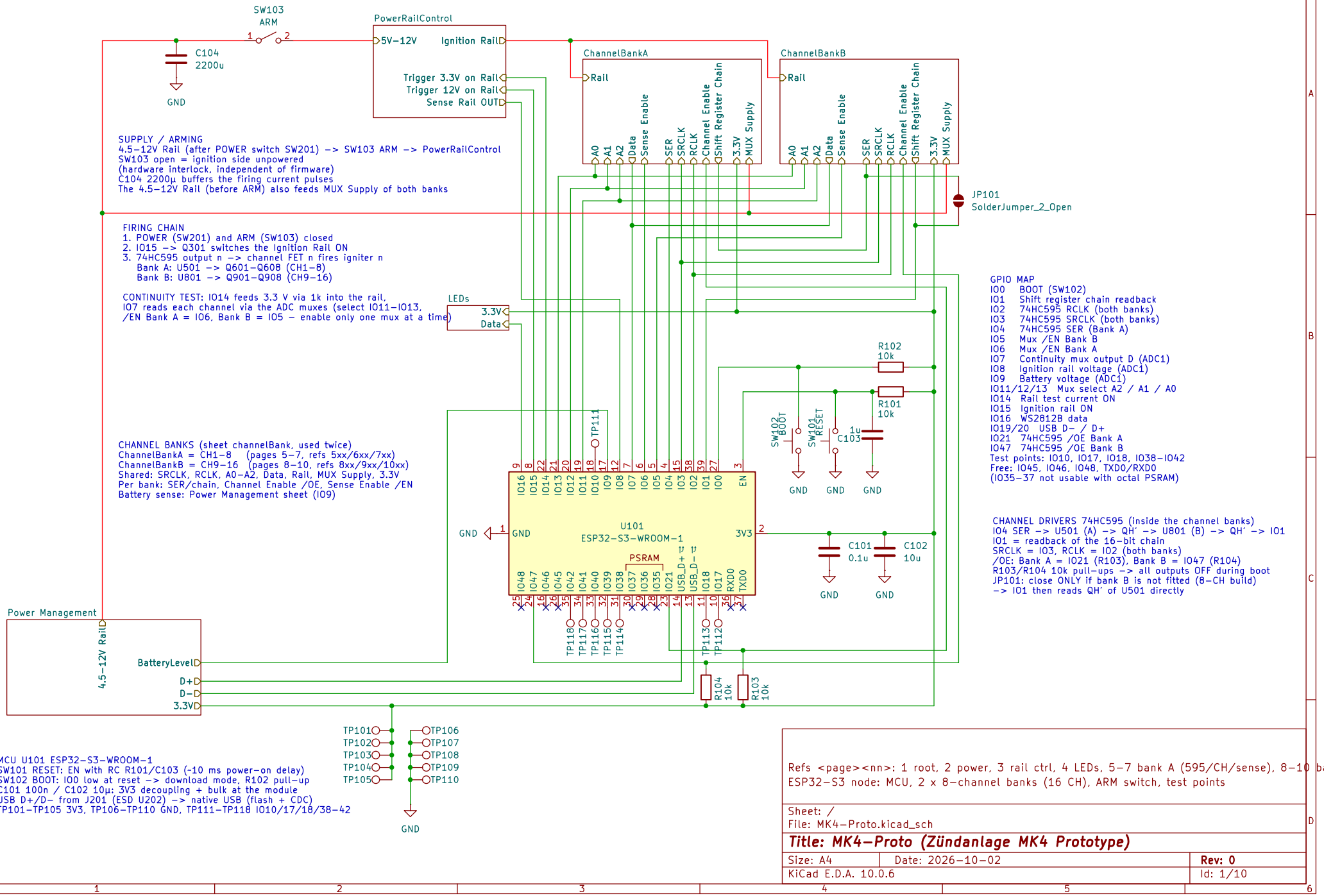
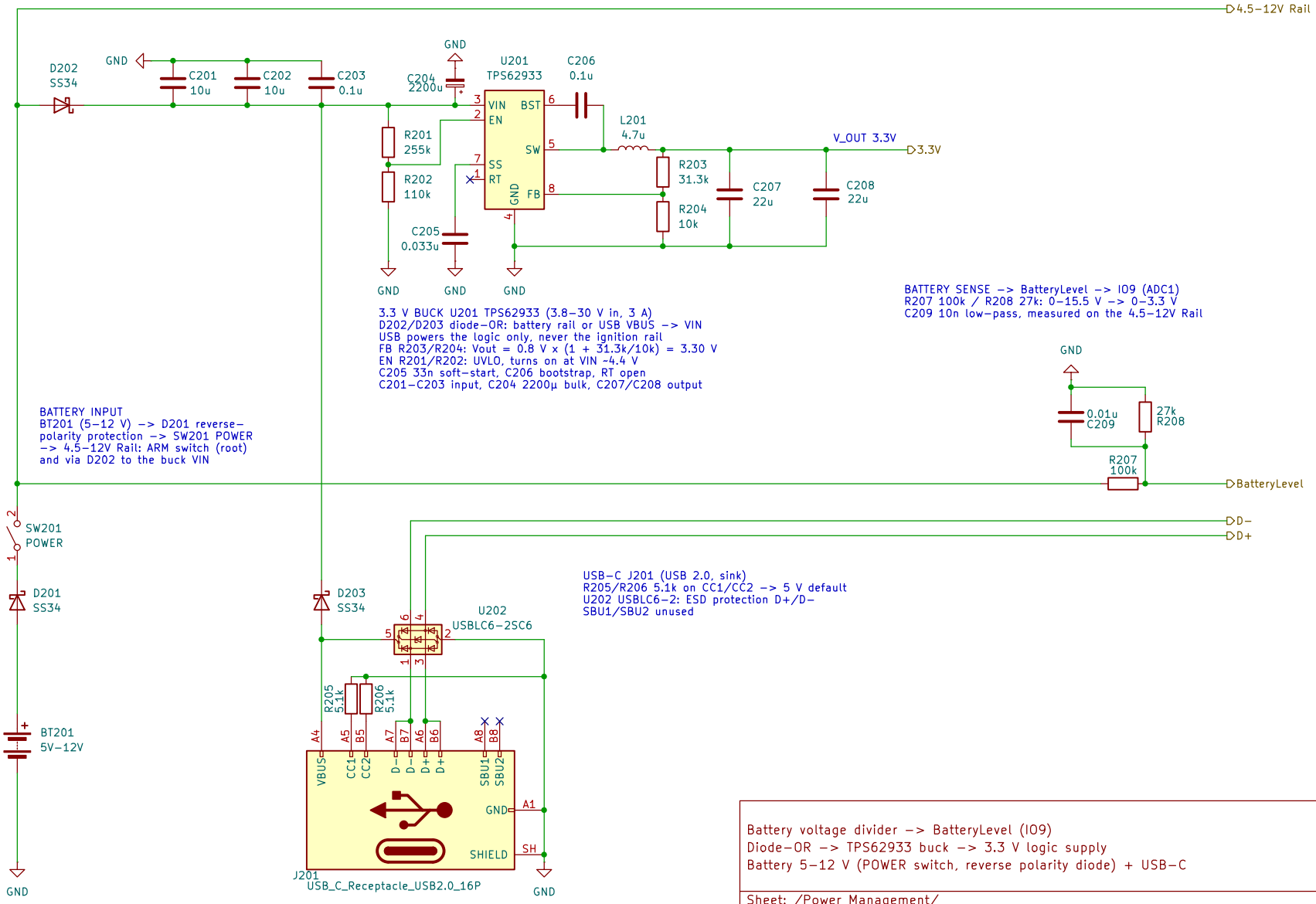
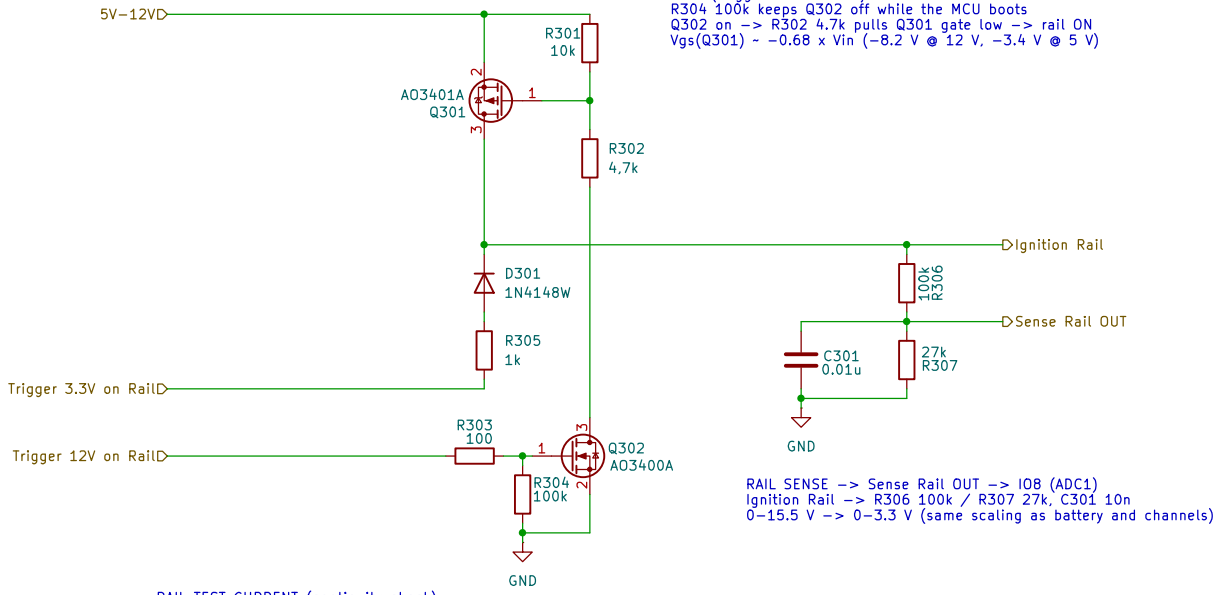

SW103
ARM
PowerRailControl
5V–12V
Ignition Rail
Trigger 3.3V on Rail
Trigger 12V on Rail
Sense Rail OUT
C104
2200u
GND
ChannelBankA
ChannelBankB
Rail
A0
A1
A2
Data
Sense Enable
SER
SRCLK
RCLK
Channel Enable
Shift Register Chain
3.3V
MUX Supply
JP101
SolderJumper_2_Open
SUPPLY / ARMING
4.5–12V Rail (after POWER switch SW201) –> SW103 ARM –> PowerRailControl
SW103 open = ignition side unpowered
(hardware interlock, independent of firmware)
C104 2200µ buffers the firing current pulses
The 4.5–12V Rail (before ARM) also feeds MUX Supply of both banks
FIRING CHAIN
1. POWER (SW201) and ARM (SW103) closed
2. IO15 –> Q301 switches the Ignition Rail ON
3. 74HC595 output n –> channel FET n fires igniter n
Bank A: U501 –> Q601–Q608 (CH1–8)
Bank B: U801 –> Q901–Q908 (CH9–16)
CONTINUITY TEST: IO14 feeds 3.3 V via 1k into the rail,
IO7 reads each channel via the ADC muxes (select IO11–IO13,
/EN Bank A = IO6, Bank B = IO5 – enable only one mux at a time)
LEDs
3.3V
Data
GPIO MAP
IO0 BOOT (SW102)
IO1 Shift register chain readback
IO2 74HC595 RCLK (both banks)
IO3 74HC595 SRCLK (both banks)
IO4 74HC595 SER (Bank A)
IO5 Mux /EN Bank B
IO6 Mux /EN Bank A
IO7 Continuity mux output D (ADC1)
IO8 Ignition rail voltage (ADC1)
IO9 Battery voltage (ADC1)
IO11/12/13 Mux select A2 / A1 / A0
IO14 Rail test current ON
IO15 Ignition rail ON
IO16 WS2812B data
IO19/20 USB D– / D+
IO21 74HC595 /OE Bank A
IO47 74HC595 /OE Bank B
Test points: IO10, IO17, IO18, IO38–IO42
Free: IO45, IO46, IO48, TXD0/RXD0
(IO35–37 not usable with octal PSRAM)
CHANNEL BANKS (sheet channelBank, used twice)
ChannelBankA = CH1–8 (pages 5–7, refs 5xx/6xx/7xx)
ChannelBankB = CH9–16 (pages 8–10, refs 8xx/9xx/10xx)
Shared: SRCLK, RCLK, A0–A2, Data, Rail, MUX Supply, 3.3V
Per bank: SER/chain, Channel Enable /OE, Sense Enable /EN
Battery sense: Power Management sheet (IO9)
R102
10k
R101
10k
SW102
BOOT
SW101
RESET
C103
1u
GND
U101
ESP32-S3-WROOM-1
PSRAM
3V3
EN
C101
0.1u
C102
10u
CHANNEL DRIVERS 74HC595 (inside the channel banks)
IO4 SER –> U501 (A) –> QH' –> U801 (B) –> QH' –> IO1
IO1 = readback of the 16–bit chain
SRCLK = IO3, RCLK = IO2 (both banks)
/OE: Bank A = IO21 (R103), Bank B = IO47 (R104)
R103/R104 10k pull–ups –> all outputs OFF during boot
JP101: close ONLY if bank B is not fitted (8–CH build)
–> IO1 then reads QH' of U501 directly
Power Management
4.5–12V Rail
BatteryLevel
D+
D–
3.3V
R104
10k
R103
10k
TP101
TP102
TP103
TP104
TP105
TP106
TP107
TP108
TP109
TP110
MCU U101 ESP32–S3–WROOM–1
SW101 RESET: EN with RC R101/C103 (~10 ms power–on delay)
SW102 BOOT: IO0 low at reset –> download mode, R102 pull–up
C101 100n / C102 10µ: 3V3 decoupling + bulk at the module
USB D+/D– from J201 (ESD U202) –> native USB (flash + CDC)
TP101–TP105 3V3, TP106–TP110 GND, TP111–TP118 IO10/17/18/38–42
Refs <page><nn>: 1 root, 2 power, 3 rail ctrl, 4 LEDs, 5–7 bank A (595/CH/sense), 8–10 bank B
ESP32–S3 node: MCU, 2 x 8–channel banks (16 CH), ARM switch, test points
Sheet: /
File: MK4–Proto.kicad_sch
Title: MK4–Proto (Zündanlage MK4 Prototype)
Size: A4
Date: 2026–10–02
Rev: 0
KiCad E.D.A. 10.0.6
Id: 1/10


4.5–12V Rail
D202
SS34
GND
C201
10u
C202
10u
C203
0.1u
C204
2200u
U201
TPS62933
C206
0.1u
VIN
EN
SS
RT
BST
SW
GND
FB
L201
4.7u
V_OUT 3.3V
3.3V
R201
255k
R202
110k
R203
31.3k
R204
10k
C205
0.033u
C207
22u
C208
22u
3.3 V BUCK U201 TPS62933 (3.8–30 V in, 3 A)
D202/D203 diode-OR: battery rail or USB VBUS -> VIN
USB powers the logic only, never the ignition rail
FB R203/R204: Vout = 0.8 V x (1 + 31.3k/10k) = 3.30 V
EN R201/R202: UVLO, turns on at VIN ~4.4 V
C205 33n soft-start, C206 bootstrap, RT open
C201–C203 input, C204 2200µ bulk, C207/C208 output
BATTERY SENSE -> BatteryLevel -> IO9 (ADC1)
R207 100k / R208 27k: 0–15.5 V -> 0–3.3 V
C209 10n low-pass, measured on the 4.5–12V Rail
0.01u
C209
27k
R208
R207
100k
BatteryLevel
BATTERY INPUT
BT201 (5–12 V) -> D201 reverse-polarity protection -> SW201 POWER -> 4.5–12V Rail: ARM switch (root) and via D202 to the buck VIN
SW201
POWER
D201
SS34
BT201
5V–12V
D203
SS34
U202
USBLC6-2SC6
D-
D+
USB-C J201 (USB 2.0, sink)
R205/R206 5.1k on CC1/CC2 -> 5 V default
U202 USBLC6-2: ESD protection D+/D-
SBU1/SBU2 unused
R205
5.1k
R206
5.1k
VBUS
CC1
CC2
SBU1
SBU2
SHIELD
SH
A1
J201
USB_C_Receptacle_USB2.0_16P
Battery voltage divider -> BatteryLevel (IO9)
Diode-OR -> TPS62933 buck -> 3.3 V logic supply
Battery 5–12 V (POWER switch, reverse polarity diode) + USB-C
Sheet: /Power Management/
File: PowerManagement.kicad_sch
Title: Power Management
Size: A4
Date: 2026-10-30
Rev:
KiCad E.D.A. 10.0.6
Id: 2/10

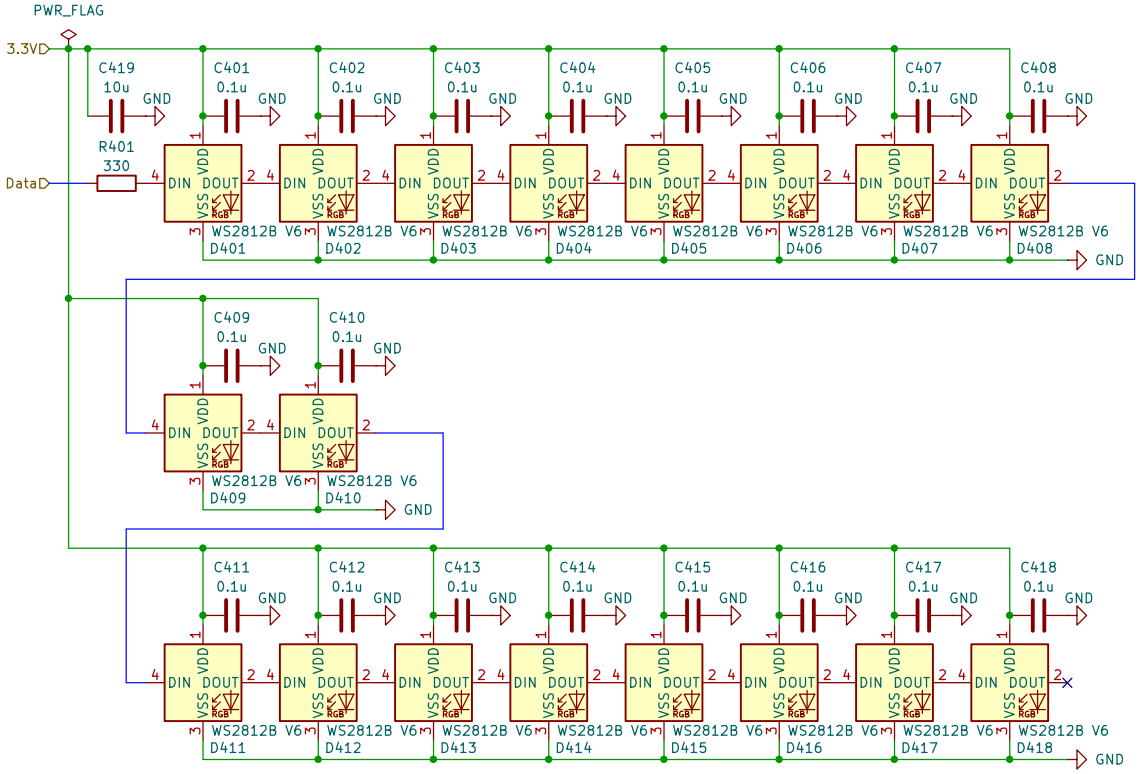


Rail voltage divider -> Sense Rail OUT (IO8)
Current-limited 3.3 V test feed for continuity check (IO14)
High-side P-FET switch for the ignition rail (IO15)

Sheet: /PowerRailControl/
File: testing.kicad_sch

Title: Ignition & Test Rail Control

Size: A4	Date: 2026-10-24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/10



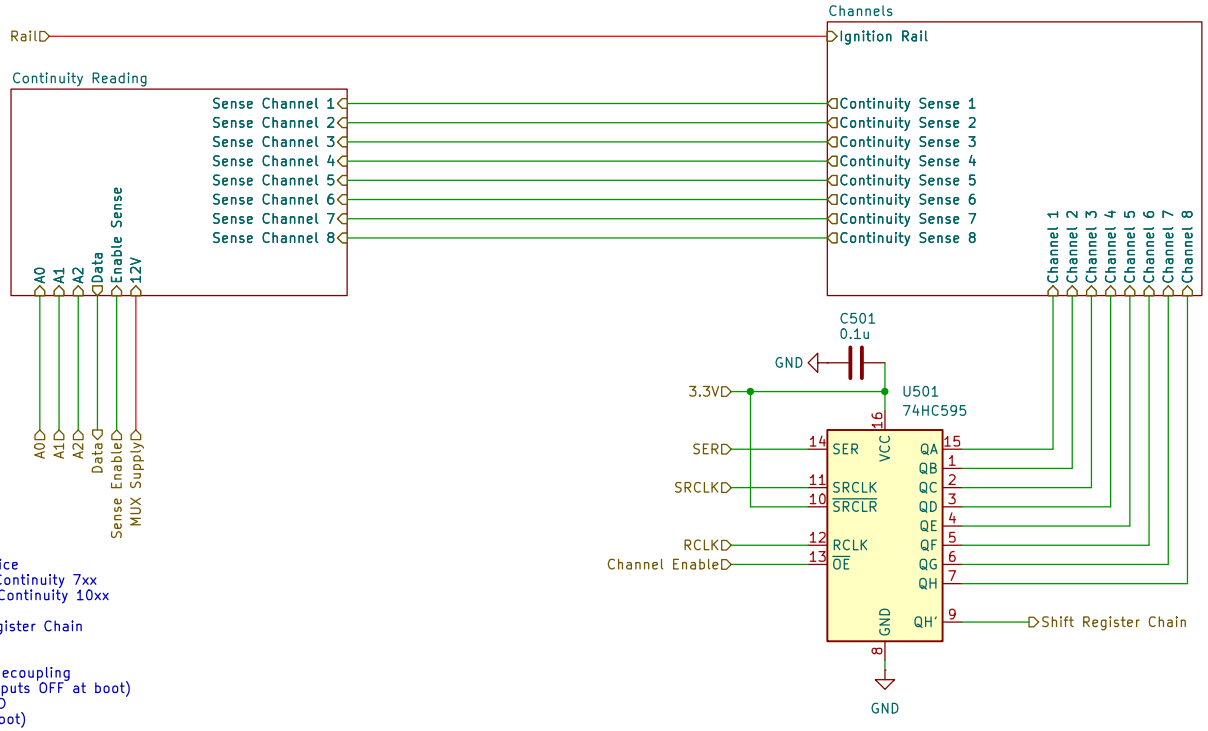
STATUS / CHANNEL LEDs
IO16 -> R401 330R series resistor -> WS2812B chain D401 ... D418
Chain order = LED index in firmware (D401 = index 0, D418 = index 17)
Row 1: D401–D408 (index 0–7)
Row 2: D409/D410 (index 8–9, status)
Row 3: D411–D418 (index 10–17)
C401–C418: 100n decoupling, one per LED, C419 10µ bulk

WS2812B chain, data from IO16
2 Status Leds
16 Channel Leds (2 x 8)

Sheet: /LEDs/
File: leds.kicad_sch

Title: LEDs

Size: A4	Date: 2026-10-24	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/10



CHANNEL BANK (8 channels) — this sheet is used twice
ChannelBankA: CH1–8, U501/C501, Channels 6xx, Continuity 7xx
ChannelBankB: CH9–16, U801/C801, Channels 9xx, Continuity 10xx

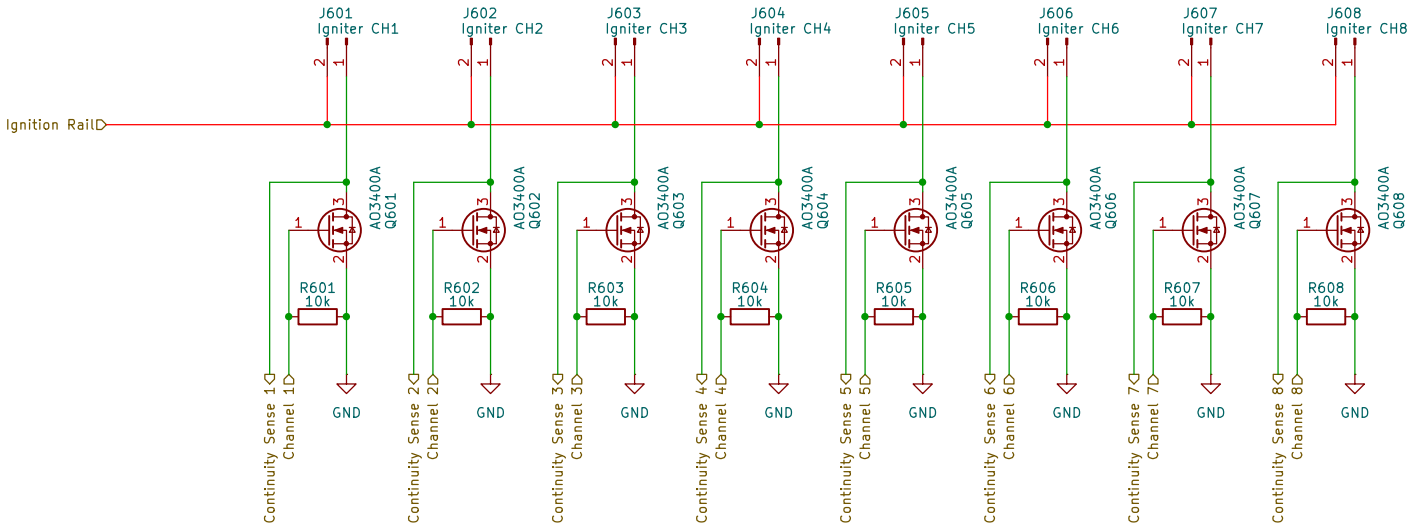
74HC595: QA..QH -> Channel 1..8, QH' -> Shift Register Chain
  Bank A: SER = IO4, chain -> SER of Bank B
  Bank B: chain -> IO1 (readback)
SRCLK / RCLK shared, /SRCLR tied to 3.3 V, 100n decoupling
/OE = Channel Enable (10k pull-up on root -> outputs OFF at boot)
Sense Enable -> mux /EN, MUX Supply -> mux VDD
8-channel build: fit only bank A and close JP101 (root)

Used twice:
ChannelBankA = CH1–8 (refs 5xx–7xx)
ChannelBankB = CH9–16 (refs 8xx–10xx)
8-channel bank: 74HC595 driver + Channels + Continuity Reading

Sheet: /ChannelBankA/
File: channelBank.kicad_sch

Title: Channel Bank

Size: A4	Date: 2026-11-01	Rev:
KiCad E.D.A. 10.0.6		Id: 5/10



IGNITION CHANNELS 1–8 of this bank (low-side switches)
Bank A = CH1–8: J601–J608, Q601–Q608, R601–R608
Bank B = CH9–16: J901–J908, Q901–Q908, R901–R908
Terminal: Pin 2 = Ignition Rail, Pin 1 = igniter return
AO3400A N-FET pulls the igniter to GND when 'Channel n' is high
10k gate pull-down keeps the channel OFF while the driver is undriven
Drain node = 'Continuity Sense n' -> divider + mux (Continuity Reading)
An igniter only fires if the Ignition Rail is ON (Q301) AND its FET is ON
Values 'Igniter CH1–8' = channel number within the bank

Current measurement line – Continuity Sense
Channel On/Off control by Mosfets
One ignition current rail

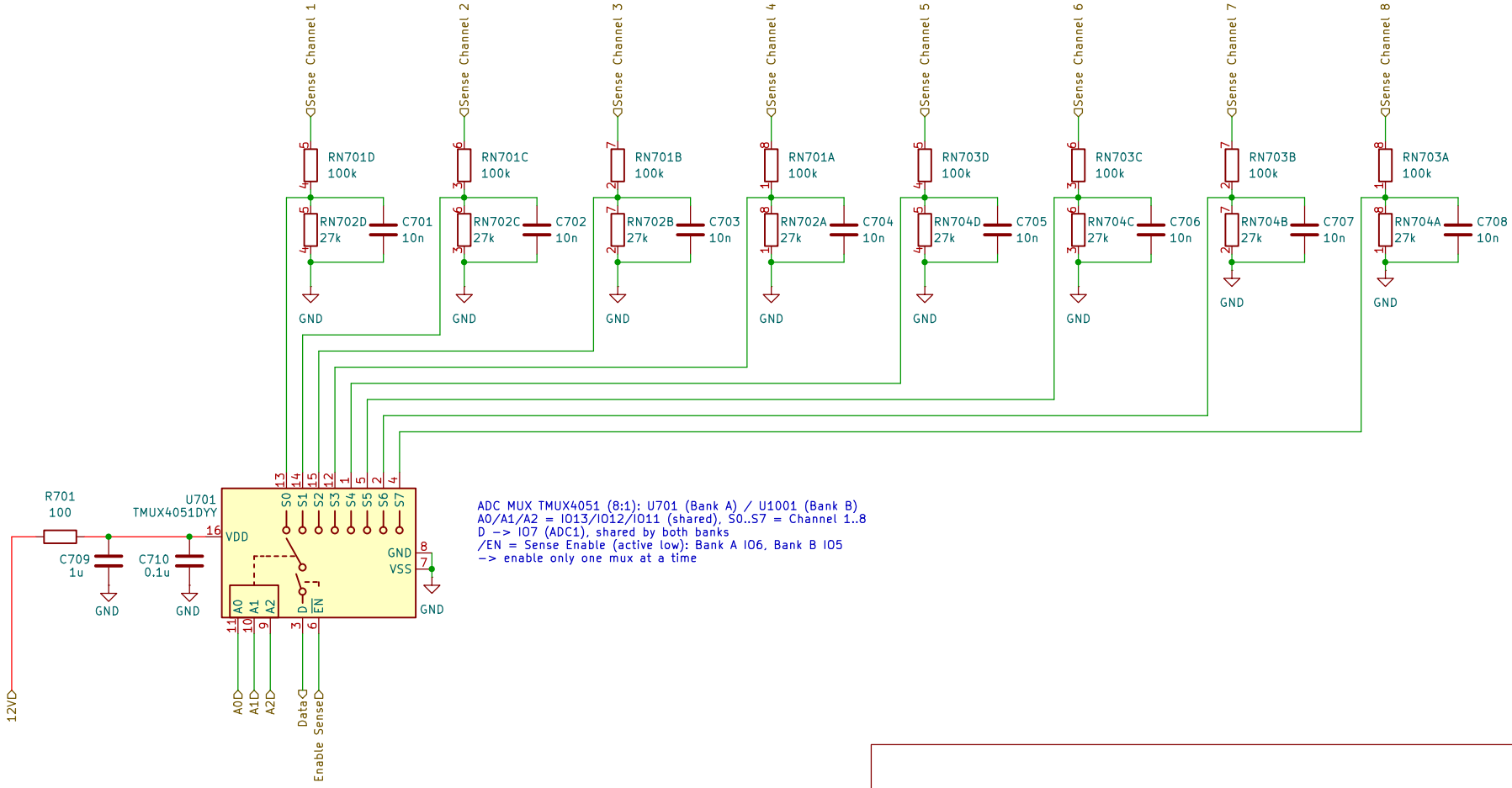
Sheet: /ChannelBankA/Channels/
File: channels.kicad_sch

Title: Channels

Size: A4	Date: 2026-10-25	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/10

CHANNEL SENSE (one per channel)
Sense Channel n -> 100k / 27k divider + 10n -> mux input S(n-1)
Bank A: Ch1-4 RN701 (100k) / RN702 (27k), Ch5-8 RN703 / RN704, C701-C708
Bank B: Ch1-4 RN1001 / RN1002, Ch5-8 RN1003 / RN1004, C1001-C1008
Igniter connected -> node follows the rail voltage, open -> ~0 V

MUX SUPPLY: MUX Supply (4.5-12V Rail, before ARM) -> R701 / R1001 100R -> VDD
C709 / C1009 1µ + C710 / C1010 100n at VDD (TMUX4051: VDD 5-24 V)



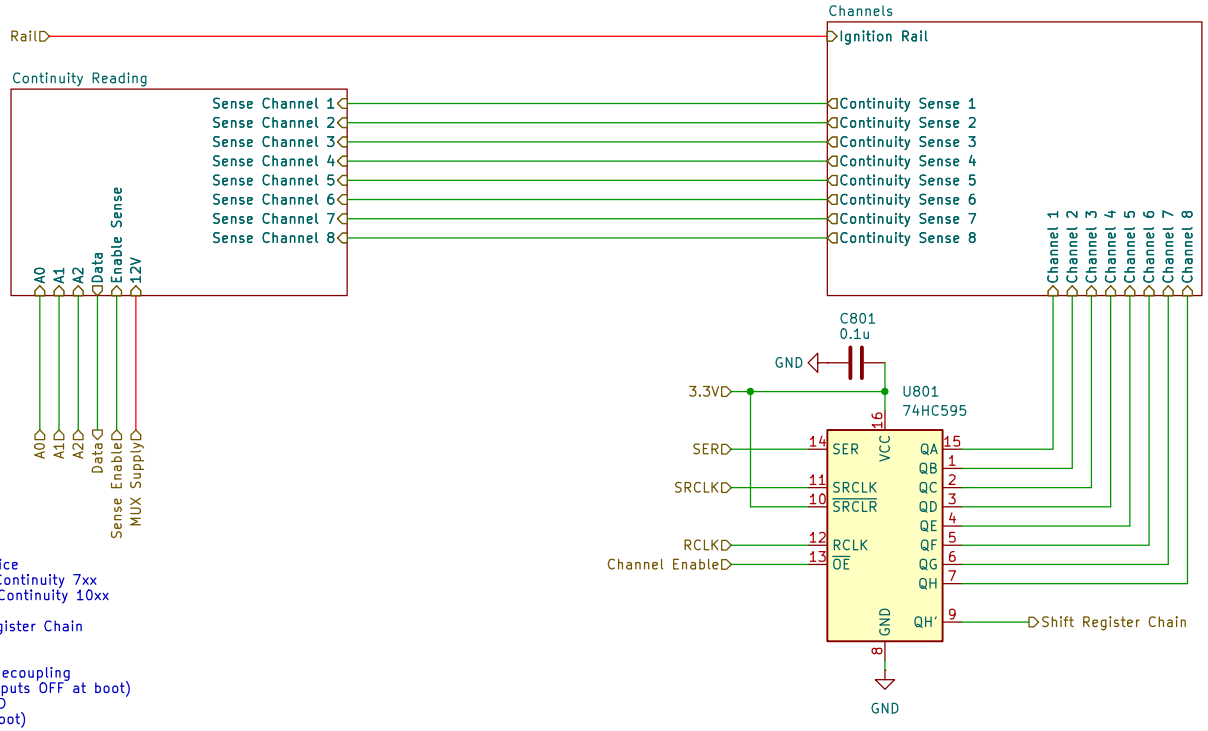
ADC MUX TMUX4051 (8:1): U701 (Bank A) / U1001 (Bank B)
A0/A1/A2 = IO13/IO12/IO11 (shared), S0..S7 = Channel 1..8
D -> IO7 (ADC1), shared by both banks
/EN = Sense Enable (active low): Bank A IO6, Bank B IO5
-> enable only one mux at a time

TMUX4051 8:1 mux, A0-A2 select, D -> ADC (shared), /EN per bank, VDD = MUX Supply
8-CH VOLTAGE SENSE: 100k/27k divider (0-15.5 V -> 0-3.3 V), 10 nF filter (fc ≈ 750 Hz)

Sheet: /ChannelBankA/Continuity Reading/
File: ContinuityReading.kicad_sch

Title: Continuity Sensing

Size: A4	Date: 2026-10-26	Rev:
KiCad E.D.A. 10.0.6		Id: 7/10



CHANNEL BANK (8 channels) — this sheet is used twice
ChannelBankA: CH1–8, U501/C501, Channels 6xx, Continuity 7xx
ChannelBankB: CH9–16, U801/C801, Channels 9xx, Continuity 10xx

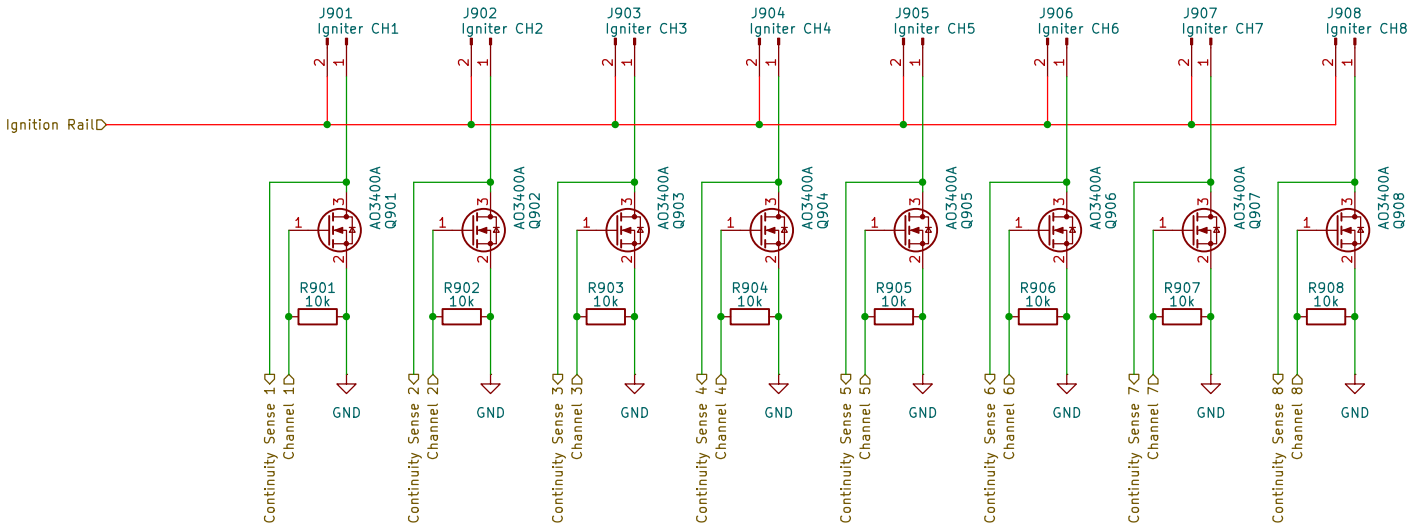
74HC595: QA..QH -> Channel 1..8, QH' -> Shift Register Chain
  Bank A: SER = IO4, chain -> SER of Bank B
  Bank B: chain -> IO1 (readback)
SRCLK / RCLK shared, /SRCLR tied to 3.3 V, 100n decoupling
/OE = Channel Enable (10k pull-up on root -> outputs OFF at boot)
Sense Enable -> mux /EN, MUX Supply -> mux VDD
8-channel build: fit only bank A and close JP101 (root)

Used twice:
ChannelBankA = CH1–8 (refs 5xx–7xx)
ChannelBankB = CH9–16 (refs 8xx–10xx)
8-channel bank: 74HC595 driver + Channels + Continuity Reading

Sheet: /ChannelBankB/
File: channelBank.kicad_sch

Title: Channel Bank

Size: A4	Date: 2026-11-01	Rev:
KiCad E.D.A. 10.0.6		Id: 8/10



IGNITION CHANNELS 1–8 of this bank (low-side switches)
Bank A = CH1–8: J601–J608, Q601–Q608, R601–R608
Bank B = CH9–16: J901–J908, Q901–Q908, R901–R908
Terminal: Pin 2 = Ignition Rail, Pin 1 = igniter return
AO3400A N-FET pulls the igniter to GND when 'Channel n' is high
10k gate pull-down keeps the channel OFF while the driver is undriven
Drain node = 'Continuity Sense n' -> divider + mux (Continuity Reading)
An igniter only fires if the Ignition Rail is ON (Q301) AND its FET is ON
Values 'Igniter CH1–8' = channel number within the bank

Current measurement line – Continuity Sense
Channel On/Off control by Mosfets
One ignition current rail

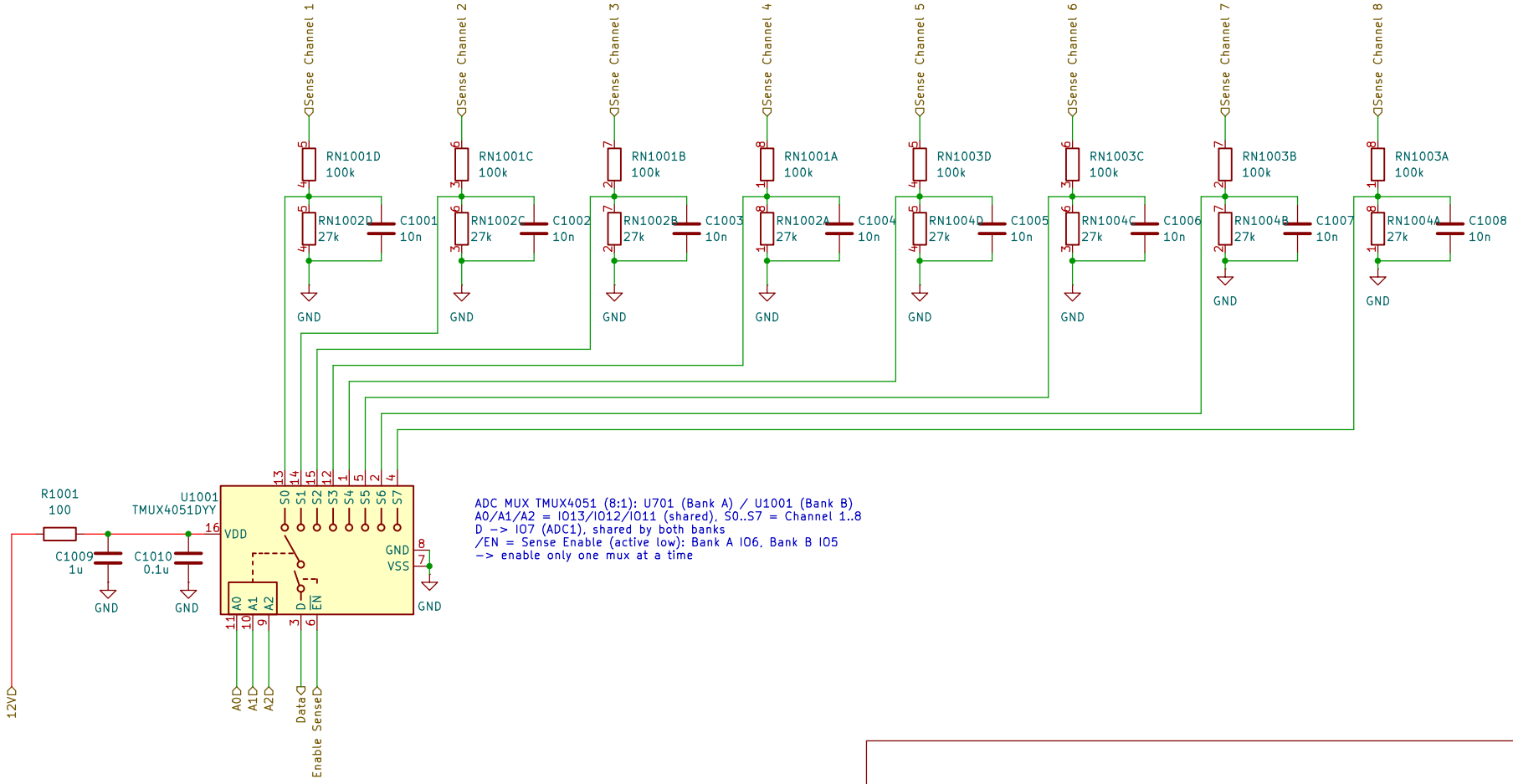
Sheet: /ChannelBankB/Channels/
File: channels.kicad_sch

Title: Channels

Size: A4	Date: 2026-10-25	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/10

CHANNEL SENSE (one per channel)
Sense Channel n -> 100k / 27k divider + 10n -> mux input S(n-1)
Bank A: Ch1-4 RN701 (100k) / RN702 (27k), Ch5-8 RN703 / RN704, C701-C708
Bank B: Ch1-4 RN1001 / RN1002, Ch5-8 RN1003 / RN1004, C1001-C1008
Igniter connected -> node follows the rail voltage, open -> ~0 V

MUX SUPPLY: MUX Supply (4.5-12V Rail, before ARM) -> R701 / R1001 100R -> VDD
C709 / C1009 1µ + C710 / C1010 100n at VDD (TMUX4051: VDD 5-24 V)



ADC MUX TMUX4051 (8:1): U701 (Bank A) / U1001 (Bank B)
A0/A1/A2 = IO13/IO12/IO11 (shared), S0..S7 = Channel 1..8
D -> IO7 (ADC1), shared by both banks
/EN = Sense Enable (active low): Bank A IO6, Bank B IO5
-> enable only one mux at a time

TMUX4051 8:1 mux, A0-A2 select, D -> ADC (shared), /EN per bank, VDD = MUX Supply
8-CH VOLTAGE SENSE: 100k/27k divider (0-15.5 V -> 0-3.3 V), 10 nF filter (fc ≈ 750 Hz)

Sheet: /ChannelBankB/Continuity Reading/
File: ContinuityReading.kicad_sch

Title: Continuity Sensing

Size: A4	Date: 2026-10-26	Rev:
KiCad E.D.A. 10.0.6		Id: 10/10